Example

Write the K_p and K_c for the following reactions

1. $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

2. $2CO(g) \rightleftharpoons CO_2(g) + C(s)$

1) $$K_C = \frac{[SO_3]^2}{[SO_2]^2\,[O_2]}$$

$$K_P = \frac{P^2_{SO_3}}{P^2_{SO_2} \times P_{O_2}}$$

2) $$K_C = \frac{[CO_2]}{[CO]^2} \quad \text{and} \quad K_P = \frac{P_{CO_2}}{P^2_{CO}}$$

Evaluate Yourself

1) Consider the following reaction

$$\mathbf{Fe^{3+}(aq) + SCN^{-}(aq) \rightleftharpoons [Fe(SCN)]^{2+}(aq)}$$

A solution is made with initial Fe^{3+}, SCN^- concentration of 1×10^{-3}M and 8×10^{-4} M respectively. At equilibrium $[Fe(SCN)]^{2+}$ concentration is 2×10^{-4}M. Calculate the value of equilibrium constant.

2) The atmospheric oxidation of NO

$$\mathbf{2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)}$$

was studied with initial pressure of 1 atm of NO and 1 atm of O_2. At equilibrium, partial pressure of oxygen is 0.52 atm calculate K_p of the reaction.

8.7 Application of equilibrium constant

The knowledge of equilibrium constant helps us to

1. predict the direction in which the net reaction will take place
2. predict the extent of the reaction and
3. calculate the equilibrium concentrations of the reactants and products.

It is to be noted that these constants do not provide any information regarding the rates of the forward or reverse reactions.

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